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BENDARED

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BENDARED

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Dr.Reddy's



Prescribing Information

BENDARED

(Bendamustine Hydrochloride concentrate for solution for infusion 180mg/4 mL)

COMPOSITION

Each vial contains Bendamustine Hydrochloride 180.0 mg/4mL.

CLINICAL INFORMATION

Therapeutic indications

Bendared is indicated for monotherapy in patients with chronic lymphocytic leukaemia. Efficacy relative to first line therapies other than chlorambucil has not been established. Bendared is indicated for monotherapy in patients with indolent B-cell non-Hodgkin's lymphomas that has progressed during or within six months of treatment with rituximab or a rituximab-containing regimen

Posology and method of administration

For intravenous infusion over 30 - 60 minutes. (see Instructions for Use and Handling and Disposal).
Infusion must be administered under the supervision of a physician qualified and experienced in the use of chemotherapeutic agents.
Poor bone marrow function is related to increased chemotherapy-induced haematological toxicity. Treatment should not be started if leukocyte and/or platelet values have dropped to < 3,000/µl or < 75,000/µl, respectively (see Contraindications).
Monotherapy for chronic lymphocytic leukaemia
100 mg/m² body surface area bendamustine hydrochloride on days 1 and 2; every 4 weeks.
Monotherapy for indolent non-Hodgkin's lymphomas refractory to rituximab
120 mg/m² body surface area bendamustine hydrochloride on days 1 and 2; every 3 weeks.
Treatment should be terminated or delayed if leukocyte and/or platelet values drop to < 3000/µL or < 75000/µL, respectively. Treatment can be continued after leukocyte values have increased to > 4000/µL and platelet values to > 100000/µL.
The leukocyte and platelet nadir is reached after 14-20 days with regeneration after 3-5 weeks. During therapy free intervals strict monitoring of the blood count is recommended (see Warnings and Precautions).
In case of non-hematological toxicity, dose reductions have to be based on the worst common toxicity criteria (CTC) grades in the preceding cycle. A 50% dose reduction is recommended in case of CTC grade 3 toxicity. An interruption of treatment is recommended in case of CTC grade 4 toxicity.
If a patient requires a dose modification, the individually calculated reduced dose must be given on day 1 and 2 of the respective treatment cycle.
For preparation and administration instructions, see Instructions for Use and Handling.

Hepatic impairment
On the basis of pharmacokinetic data, no dose adjustment is necessary in patients with mild hepatic impairment (serum bilirubin < 1.2 mg/dL). A 30% dose reduction is recommended in patients with moderate hepatic impairment (serum bilirubin 1.2 - 3.0 mg/dL).
No data are available in patients with severe hepatic impairment (serum bilirubin values of > 3.0 mg/dL) (see Contraindications).
Renal impairment
On the basis of pharmacokinetic data, no dose adjustment is necessary in patients with a creatinine clearance of > 10 mL/min. Experience in patients with severe renal impairment is limited.
Pediatric patients
As there are limited data, the safety and efficacy of bendamustine in pediatric patients has not been established.
Elderly patients
There is no evidence that dose adjustments are necessary in elderly patients (see Pharmacokinetic Properties).
Method of administration
For intravenous infusion over 30 - 60 minutes
Infusion must be administered under the supervision of a physician qualified and experienced in the use of chemotherapeutic agents.
Precautions to be taken before handling or administering the medicinal product.

Contraindications

- Hypersensitivity to the active substance or to any of the excipients

- During breastfeeding

- Severe hepatic impairment (serum bilirubin > 3.0 mg/dl)

- Jaundice

- Severe bone marrow suppression and severe blood count alterations (leukocyte and/or platelet values dropped to < 3,000/µl or < 75,000/µl, respectively)

- Major surgery less than 30 days before start of treatment

- Infections, especially involving leukocytopenia

- Yellow fever vaccination

Special warnings and precautions for use

Myelosuppression
Patients treated with bendamustine hydrochloride may experience myelosuppression. In the event of treatment-related myelosuppression, leukocytes, platelets, haemoglobin, and neutrophils must be monitored at least weekly. Prior to the initiation of the next cycle of therapy, the following parameters are recommended: Leukocyte and/or platelet values > 4,000/µl or > 100,000/µl, respectively.
Infections
Serious and fatal infections have occurred with bendamustine hydrochloride, including bacterial (sepsis, pneumonia) and opportunistic infections such as Pneumocystis jirovecii pneumonia (PJP), varicella zoster virus (VZV) and cytomegalovirus (CMV). Treatment with bendamustine hydrochloride may cause prolonged lymphocytopenia (< 600/µl) and low CD4-positive T-cell (T-helper cell) counts (< 200/µl) for at least 7-9 months after the completion of treatment. Lymphocytopenia and CD4-positive T-cell depletion are more pronounced when bendamustine is combined with rituximab. Patients with lymphopenia and low CD4-positive T-cell count following treatment with bendamustine hydrochloride are more susceptible to (opportunistic) infections. In case of low CD4-positive T-cell counts (< 200 µl) Pneumocystis jirovecii pneumonia (PJP) prophylaxis should be considered. All patients should be monitored for respiratory signs and symptoms throughout treatment. Patients should be advised to report new signs of infection, including fever or respiratory symptoms promptly. Discontinuation of bendamustine hydrochloride should be considered if there are signs of (opportunistic) infections.
Hepatitis B reactivation
Reactivation of hepatitis B in patients who are chronic carriers of this virus has occurred after these patients received bendamustine hydrochloride. Some cases resulted in acute hepatic failure or a fatal outcome. Patients should be tested for HBV infection before initiating treatment with bendamustine hydrochloride. Experts in liver disease and in the treatment of hepatitis B should be consulted before treatment is initiated in patients with positive hepatitis B tests (including those with active disease) and for patients who test positive for HBV infection during treatment. Carriers of HBV who require treatment with bendamustine hydrochloride should be closely monitored for signs and symptoms of active HBV infection throughout therapy and for several months following termination of therapy.
Skin reactions
A number of skin reactions have been reported. These events have included rash, severe cutaneous reactions and bullous exanthema. Cases of Stevens – Johnson syndrome (SJS) and Toxic Epidermal Necrolysis (TEN), and Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS), some fatal, have been reported with the use of bendamustine hydrochloride. Patients should be advised of the signs and symptoms of these reactions by their prescribers and should be told to seek medical attention immediately if they develop these symptoms. Some events occurred when bendamustine hydrochloride was given in combination with other anticancer agents, so the precise relationship is uncertain. When skin reactions occur, they may be progressive and increase in severity with further treatment. If skin reactions are progressive, bendamustine should be withheld or discontinued. For severe skin reactions with suspected relationship to bendamustine hydrochloride, treatment should be discontinued.
Cardiac disorders
During treatment with bendamustine hydrochloride the concentration of potassium in the blood of patients with cardiac disorders must be closely monitored and potassium supplement must be given when K⁺ <3.5 mEq/l and ECG measurement must be performed.
Fatal cases of myocardial infarction and cardiac failure have been reported with bendamustine hydrochloride treatment. Patients with concurrent or history of cardiac disease should be observed closely.
Nausea, vomiting
An antiemetic may be given for the symptomatic treatment of nausea and vomiting.
Tumour lysis syndrome
Tumour lysis syndrome (TLS) associated with bendamustine treatment has been reported in patients. The onset tends to be within 48 hours of the first dose of bendamustine and, without intervention, may lead to acute renal failure and death. Preventive measures such as adequate hydration, close monitoring of blood chemistry, particularly potassium and uric acid levels and the use of hypouricemic agents (allopurinol and rasburicase) should be considered prior to therapy. There have been a few cases of Stevens-Johnson Syndrome and Toxic Epidermal Necrolysis reported when bendamustine and allopurinol were administered concomitantly.
Anaphylaxis
Infusion reactions to bendamustine hydrochloride have occurred commonly. Symptoms are generally mild and include fever, chills, pruritus and rash. In rare instances severe anaphylactic and anaphylactoid reactions have occurred. Patients must be asked about symptoms suggestive of infusion reactions after their first cycle of therapy. Measures to prevent severe reactions, including antihistamines, antipyretics and corticosteroids must be considered in subsequent cycles in patients who have previously experienced infusion reactions.
Patients who experienced Grade 3 or worse allergic-type reactions were typically not re-challenged.
Contraception
Bendamustine hydrochloride is teratogenic and mutagenic.
Women should not become pregnant during treatment. Male patients should not father a child during and up to 6 months after treatment. They should seek advice about sperm conservation prior to treatment with bendamustine hydrochloride because of possible irreversible infertility.
Extravasation
An extravasal injection should be stopped immediately. The needle should be removed after a short aspiration. Thereafter the affected area of tissue should be cooled. The arm should be elevated. Additional treatments like the use of corticosteroids are not of clear benefit.
Dilution
Bendamustine requires appropriate dilution before use. The concentration of bendamustine in Bendamustine differs from other bendamustine products.

Interaction with other medicinal products and other forms of interaction

No interaction data is available.
When bendamustine is combined with myelosuppressive agents, the effect of bendamustine and/or the co-administered medicinal products on the bone marrow may be potentiated. Any treatment reducing the patient's performance status or impairing bone marrow function can increase the toxicity of bendamustine.
Combination of bendamustine with cyclosporine or tacrolimus may result in excessive immunosuppression with risk of lymphoproliferation.
Cytostatics can reduce antibody formation following live-virus vaccination and increase the risk of infection which may lead to fatal outcome. This risk is increased in patients who are already immunosuppressed by their underlying disease.
Bendamustine metabolism involves cytochrome P450 (CYP) 1A2 isoenzyme. Therefore, the potential for interaction with CYP1A2 inhibitors such as fluvoxamine, ciprofloxacin, acyclovir and cimetidine exists.

Paediatric population

Interactions have only been performed in adults.

Fertility, pregnancy and lactation

Pregnancy
There are insufficient data from the use of bendamustine in pregnant women. During pregnancy bendamustine should not be used unless clearly necessary. The mother should be informed about the risk to the foetus. If treatment with bendamustine is absolutely necessary during pregnancy or if pregnancy occurs during treatment, the patient should be informed about the risks for the unborn child and be monitored carefully. The possibility of genetic counselling should be considered.
Fertility
Women of childbearing potential must use effective methods of contraception both before and during bendamustine therapy.
Men being treated with bendamustine are advised not to father a child during and for up to 6 months following cessation of treatment. Advice on conservation of sperm should be sought prior to treatment because of the possibility of irreversible infertility due to therapy with bendamustine.
Breast feeding
It is not known whether bendamustine passes into the breast milk, therefore, bendamustine is contraindicated during breast feeding. Breast feeding must be discontinued during treatment with bendamustine.

Effects on ability to drive and use machines

Bendamustine has major influence on the ability to drive and use machines. Ataxia, peripheral neuropathy and somnolence have been reported during treatment with bendamustine. Patients should be instructed that if they experience these symptoms they should avoid potentially hazardous tasks such as driving and using machines.

Undesirable effects

Summary of safety profile
The most common adverse reactions with bendamustine hydrochloride are hematological adverse reactions (leukopenia, thrombopenia), dermatologic toxicities (allergic reactions), constitutional symptoms (fever), gastrointestinal symptoms (nausea, vomiting).
Tabulated list of adverse reactions
The table below reflects the data obtained with bendamustine hydrochloride.

MedDRA system organ class	Very common	Common	Uncommon	Rare	Very rare	Not known
Infections and infestations	Infection NOS* Including Opportunistic infection (e.g. Herpes zoster, cytomegalovirus, hepatitis B)		Pneumocystis jirovecii pneumonia	Sepsis	Pneumonia primary atypical	
Neoplasms benign, malignant		Tumour lysis syndrome	Myelodysplastic syndrome, acute myeloid leukemia			
Blood and lymphatic system disorders	Leukopenia NOS*, Thrombocytopenia Lymphopenia	Haemorrhage Anaemia, Neutropenia	Pancytopenia	Bone marrow failure	Haemolysis	
Immune system disorders		Hypersensitivity NOS*		Anaphylactic reaction, Anaphylactoid reaction	Anaphylactic shock	
Nervous system disorders	Headache	Insomnia Dizziness		Somnolence, Aponhia	Dysgeusia, Paraesthesia, Peripheral sensory neuropathy, Anticholinergic syndrome, Neurological disorders, Ataxia, Encephalitis	

Page 1 of 2

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Cardiac disorders		Cardiac dysfunction, such as palpitations, angina pectoris, Arrhythmia	Pericardial effusion Myocardial infarction, Cardiac failure		Tachycardia	Atrial fibrillation
Vascular disorders		Hypotension, Hypertension		Acute circulatory failure	Phlebitis	
Respiratory, thoracic and media-stinal disorders		Pulmonary dysfunction			Pulmonary fibrosis	Pneumonitis, Pulmonary alveolar haemorrhage
Gastrointestinal disorders	Nausea, Vomiting	Diarrhoea, Constipation, Stomatitis			haemorrhagic oesophagitis, Gastrointestinal haemorrhage	
Hepatobiliary disorder						Hepatic failure
Skin and subcutaneous tissue disorders		Alopecia, Skin disorders NOS* Urticaria		Erythema, Dermatitis, Pruritus, Maculopapular Rash, Hyperhidrosis		Stevens – Johnson syndrome, Toxic Epidermal Necrolysis (TEN) Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS)
Renal and urinary disorders						Renal failure
Reproductive system and breast disorders		Amenorrhea			Infertility	
General disorders and administration site conditions	Mucosal inflammation, Fatigue, Pyrexia	Pain, Chills, Dehydration, Anorexia			Multi organ failure	
Investigations	Haemoglobin decrease, Creatinine increase, Urea increase	AST increase, ALT increase, Alkaline phosphatase increase, Bilirubin increase, Hypokalemia				

NOS = Not otherwise specified
(*=combination therapy with rituximab)

Description of selected adverse reactions
There have been isolated reports of necrosis after accidental extra-vascular administration and tumour lysis syndrome, and anaphylaxis. The risk of myelodysplastic syndrome and acute myeloid leukaemias is increased in patients treated with alkylating agents (including bendamustine). The secondary malignancy may develop several years after chemotherapy has been discontinued.

Overdose
After application of a 30 min infusion of bendamustine hydrochloride once every 3 weeks the maximum tolerated dose (MTD) was 280 mg/m². Cardiac events of CTC grade 2 which were compatible with ischaemic ECG changes occurred which were regarded as dose limiting. A 30 min infusion of bendamustine hydrochloride at day 1 and 2 every 3 weeks the MTD was found to be 180 mg/m². The dose limiting toxicity was grade 4 thrombocytopenia. Cardiac toxicity was not dose limiting with this schedule.

Counter measures
There is no specific antidote. Bone marrow transplantation and transfusions (platelets, concentrated erythrocytes) may be made or haematological growth factors may be given as effective countermeasures to control haematological side effects. Bendamustine hydrochloride and its metabolites are dialyzable to a small extent.

Pharmacological properties
Pharmacodynamic properties
Pharmacotherapeutic group: Antineoplastic agents, alkylating agents.
ATC code: L01AA09
Bendamustine hydrochloride is an alkylating antitumour agent with unique activity. The antineoplastic and cytotoxic effect of bendamustine hydrochloride is based essentially on a cross-linking of DNA single and double strands by alkylation. As a result, DNA matrix functions and DNA synthesis and repair are impaired. The antitumour effect of bendamustine hydrochloride has been demonstrated by several data in different human tumour cell lines (breast cancer, non-small cell and small cell lung cancer, ovarian carcinoma and different leukaemia) and in different experimental tumour models with tumours of mouse, rat and human origin (melanoma, breast cancer, sarcoma, lymphoma, leukaemia and small cell lung cancer). Bendamustine hydrochloride showed an activity profile in human tumour cell lines different to that of other alkylating agents. The active substance revealed no or very low cross-resistance in human tumour cell lines with different resistance mechanisms at least in part due to a comparatively persistent DNA interaction. Additionally, it was shown in clinical studies that there is no complete cross-resistance of bendamustine with anthracyclines, alkylating agents or rituximab. However, the number of assessed patients is small.

Pharmacokinetic properties
Distribution
The elimination half-life t_{1/2} after 30 min i.v. infusion of 120 mg/m² area was 28.2 minutes. Following 30 min i.v. infusion the central volume of distribution was 19.3 l. Under steady-state conditions following i.v. bolus injection the volume of distribution was 15.8-20.5 l. More than 95% of the substance is bound to plasma proteins (primarily albumin).
Metabolism
A major route of clearance of bendamustine is the hydrolysis to monohydroxy- and dihydroxy-bendamustine. Formation of N-desmethyl-bendamustine and gamma-hydroxy-bendamustine by hepatic metabolism involves cytochrome P450 (CYP) 1A2 isoenzyme. Another major route of bendamustine metabolism involves conjugation with glutathione. Bendamustine does not inhibit CYP 1A4, CYP 2C9/10, CYP 2D6, CYP 2E1 or CYP 3A4.
Elimination
The mean total clearance after 30 min i.v. infusion of 120 mg/m² body surface area was 639.4 ml/minute. About 20% of the administered dose was recovered in urine within 24 hours. Amounts excreted in urine were in the order monohydroxy-bendamustine > bendamustine > dihydroxy-bendamustine > oxidised metabolite > N-desmethyl bendamustine. In the bile, primarily polar metabolites are eliminated.
Hepatic impairment
In patients with 30 - 70% tumour infestation of the liver and mild hepatic impairment (serum bilirubin < 1.2 mg/dl) the pharmacokinetic behaviour was not changed. There was no significant difference to patients with normal liver and kidney function with respect to C_{max}, t_{max}, AUC, t_{1/2}, volume of distribution and clearance. AUC and total body clearance of bendamustine correlate inversely with serum bilirubin.
Renal impairment
In patients with creatinine clearance > 10 ml/min including dialysis dependent patients, no significant difference to patients with normal liver and kidney function was observed with respect to C_{max}, t_{max}, AUC, t_{1/2}, volume of distribution and clearance.
Elderly subjects
Higher age does not influence the pharmacokinetics of bendamustine.

Pharmaceutical information
Product Description
Clear, pale yellow to yellow viscous solution free from Visible extraneous matter.
Diluted Solution: A colorless solution free from visible extraneous material
Shelf Life
Unopened Vial
2 years. The concentrate should be diluted with 0.9% sodium chloride solution.
After opening of the vial
Chemical and physical in use stability has been demonstrated for 28 days at 2°–8°C.
From a microbiological point of view, once opened the product may be stored for a maximum of 28 days at 2°–8°C. Other in-use storage times and conditions are the responsibility of the user.
Minimisation of the risk of contamination of the multidose vial during withdrawal of each dose is the responsibility of the user. Record date and time of the first dose withdrawal on the vial label. Between uses, return the multidose vial to the recommended storage condition of 2°–8°C.
Solution for infusion
After dilution, chemical and physical stability has been demonstrated for 3.5 hours at 25°C/60% RH and 2 days at 2°C–8°C in polyethylene bags.
From a microbiological point of view, the solution should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user.
Storage conditions
Unopened Vial
Store and transport refrigerated (2 – 8°C). Do not freeze.
Keep the vial in the outer carton in order to protect from light.
After opening of the vial
Chemical and physical in use stability has been demonstrated for 28 days at 2°–8°C.
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Nature and contents of container
One 5ml Amber USP Type I Glass vial with grey coloured chlorobutyl coated rubber stopper and orange coloured polypropylene plastic disk with aluminium seal. Each vial placed into an outer carton with a pack insert.

Special precautions for disposal and other handling
When handling Bendamustine, inhalation, skin contact or contact with mucous membranes should be avoided (wear gloves and protective clothes!). Contaminated body parts should be carefully rinsed with water and soap, the eyes should be rinsed with physiological saline solution. If possible it is recommended to work on special safety workbenches (laminar flow) with liquid-impermeable, absorbent disposable foil. Pregnant personnel should be excluded from handling cytostatics.
The concentrate for solution for infusion has to be diluted with sodium chloride 9 mg/ml (0.9%) solution for injection and then administered by intravenous infusion. Aseptic technique is to be used.

Incompatibilities
This medicinal product must not be mixed with other medicinal products except those mentioned in below section

1. Dilution
Aseptically withdraw the volume needed for the required dose from the bendamustine 180 mg/4 ml vial. Dilute the total recommended dose of Bendamustine hydrochloride 180 mg/4 ml with 0.9% sodium chloride solution to produce a final volume of about 500 ml.
While diluting the product it should be noted that the concentration (45 mg/ml) of bendamustine in Bendamustine is higher than in usual bendamustine concentrates resulting from reconstitution of bendamustine powder containing medicinal products.
Bendamustine hydrochloride 180 mg/4 ml must be diluted with 0.9% NaCl solution and not with any other injectable solutions.

2. Administration
The solution is administered by intravenous infusion over 30-60 min.
The vials are for multiple dose use.
Any unused product or waste material should be disposed of in accordance with local requirements.

Manufactured in India by:
Dr. Reddy's Laboratories Ltd.
Formulation Unit – VII,
Plot No. P1 to P9, Phase III, Duvvada, VSEZ,
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PRODUCT REGISTRATION HOLDER
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Date of revision: February 2025